Deep Reinforcement Learning Table

Full Title	**Adaptive Neural Traffic Orchestration: AI-Driven Network Optimization for Dynamic Congestion Mitigation**	**Adaptive urban traffic signal control based on enhanced deep reinforcement learning**	**Joint Pedestrian and Vehicle Traffic Optimization in Urban Environments using Reinforcement Learning**
Publication Year	2020	2024	2025
Publisher	International Journal of Research and Analytical Reviews (IJRAR)-ResearchGate	Scientific Reports -nature	arXiv
Scientific Goal	introduces Adaptive Neural Traffic Orchestration (ANTO), a comprehensive AI-driven framework for dynamic congestion mitigation in urban transportation networks.	To propose an enhanced deep reinforcement learning–based traffic signal control (TSC) method that accelerates training convergence, alleviates the overestimation problem in DQN, and improves model robustness under diverse and dynamic urban traffic flow conditions.	Optimize both pedestrian and vehicle traffic performance jointly using RL control across a corridor of traffic signals informed by real-world demand data.
Study Focus	ANTO adaptively responds to real-time traffic conditions by orchestrating signal timing, route guidance, and demand management interventions, leveraging deep reinforcement learning and recurrent neural network architectures.	adaptive traffic signal control at a single four-leg urban intersection using an improved Deep Q-Network framework that integrates dueling network architecture, double Q-learning, prioritized experience replay, and noise injection mechanisms.	present a deep RL framework for adaptive control of eight traffic signals along a real-world urban corridor, jointly optimizing both pedestrian and vehicular efficiency.

Dataset Used	Real-world traffic datasets: -PEMS Bay Area dataset -NYC Taxi Data dataset -Manchester Traffic signal and flow dataset For simulation experiments, we used the SUMO, constructed digital twins of selected network segments from each dataset.	traffic flow dataset generated in the SUMO traffic simulation platform.	Real-world Craver Road Corridor dataset from UNC Charlotte including Wi-Fi logs and video recordings, mapped into SUMO simulation with Origin–Destination trip generation.
Data Type	Real-world traffic sensor measurements, taxi trips records, signalized intersections and road segments.	Synthetic traffic flow data generated in SUMO simulation.	Multi-modal spatio-temporal traffic data: Pedestrian mobility inferred from Wi-Fi access point logs, video-based vehicle counts, OD trip modeling using TAZ modeling.
Data Size	6 months of traffic sensor data from 325 sensors, 1.1 billion taxi trips, 79 signalized intersections and 208 road segments.	length of each vehicle is 5 m, with a maximum speed of 13.9 m per second, a maximum acceleration of 0.8 m per second squared, a deceleration of 4.5 m per second squared, and a minimum distance between two vehicles of 2 m.	88,409 unique devices, 2223 pedestrians/hr (990 crossing), 202 vehicles/hr, 750m corridor, 8 traffic signals.
Base Model	Hybrid deep learning and reinforcement learning framework combining **Graph Convolutional Networks (GCN)** for spatial traffic modeling, **LSTM-based temporal prediction**, and **Proximal Policy Optimization (PPO)** for multi-objective traffic control decision making.	DRL based on Deep Q-Network **(DQN)** with Value-based architecture, enhanced into a Dueling Double Deep Q-Network (D3QN), and further extended into the proposed **PN_D3QN** model by integrating prioritized experience replay (PER) and noisy network mechanisms.	Deep Reinforcement Learning framework for corridor-level Adaptive Traffic Signal Control (ATSC) using **Proximal Policy Optimization (PPO)** with Actor–Critic architecture.
Training Approach	Policy learning is performed using the Proximal Policy Optimization (PPO) algorithm within a reinforcement learning framework where the agent interacts with a simulated urban traffic environment. reinforcement agent optimized with multi-objective rewards balancing travel time, emissions, fairness, and stability.	The agent is trained to approximate the optimal action-value function using a convolutional and fully connected neural network. Transitions are stored as (s, a, r, s') in replay memory, and prioritized experience replay is applied for non-uniform sampling. The model integrates dueling network architecture, double Q-learning, and noise injection, with parameters updated through	Centralized training Single Agent trained using (PPO) with Generalized Advantage Estimation (GAE), Training occurs across 24 parallel actors over 6×10^6 simulation timesteps.

		gradient descent to maximize expected reward.	
Results Summary	Compared to fixed-time control, ANTO achieved a 27% reduction in average travel time, 36% drop in congestion delays, ~23% lower emissions, and improved throughput and fairness.	The enhanced DRL model converged faster than baseline DRL, and demonstrated lower average waiting times and queue lengths, showing robustness across various traffic densities.	Average pedestrian and vehicle wait times reduced (~67% and ~52% respectively) compared to fixed signals; generalization capabilities across varying traffic demands, including conditions entirely unseen during training .
Scientific Contribution	integrates real-time traffic state representation with predictive modeling capabilities , Introduction of a multi-objective reinforcement learning algorithm that balances competing traffic management goals, including travel time minimization, emissions reduction, and fairness in resource allocation. , Empirical validation of ANTO using both simulation environments and real-world traffic data from three metropolitan areas.	The model demonstrates improved performance in reducing vehicle waiting time and queue length compared to baseline methods.	Joint optimization of multimodal traffic flows and safety challenges, expanding beyond vehicle-only approaches.
Technical Innovation	ability to simultaneously coordinate multiple traffic management strategies, including signal timing, route guidance, and demand management, ANTO implements a network-wide optimization strategy that considers the complex interactions between different parts of the transportation system.	1.Integration of dueling networks and double Q-learning 2.Prioritized sampling and noise injection to accelerate learning 3.Realistic reward design aligned with queue length metrics.	EI-MWAQ exponential reward
Applications	Intelligent transportation systems, real-time adaptive traffic signal control, congestion mitigation tools for smart cities.	Adaptive traffic signal control systems; urban intersections where demands vary frequently and require fast adaptive solutions.	Urban corridors with high pedestrian movements; smart intersections serving multiple mobility classes.
Limitations / Research Gaps	Relies on high-quality real-time data; not fully tested under extreme events or	The study is limited to a single intersection scenario. no real-world deployment.	1-vehicle queue propagation upstream when signals are closely spaced

	sensor failures; computational demands may be high. current implementation not fully account for emerging mobility services like ride-sharing and autonomous vehicles.	The framework does not extend to multi-agent or network-level signal coordination. More real-world traffic data is suggested for future validation. The cold-start problem remains unresolved during early training.	2- high switching frequency, increase vehicle energy consumption 3- evaluation was limited to a university campus setting 4- baselines include common real-world approaches (fixed-time control and unsignalized crosswalks), direct comparison with other RL-based controllers proved challenging, No mixed-traffic (AV + human) modeling.
Academic Recommendation	address robustness to partial data, extreme conditions, sensor failures, multi-modal traffic, and transfer learning for deployment across varied cities.	Investigate offline DRL, safe exploration methods, and multi-agent coordination to address multi-intersection scalability and cold start issues.	explicitly modeling correlations between adjacent signals, incorporate traffic flow continuity metrics into the reward function, evaluate performance across diverse urban settings with varying pedestrian-vehicle ratios, mixed-traffic control , Assess robustness under real-world traffic disruptions such as incidents and demand fluctuations.
state representation	The state represents the current traffic conditions across the network.	The state is encoded using discrete matrix representations of vehicle positions and velocities, along with the current signal phase. Each lane is divided into grids to capture spatial vehicle distribution, forming position and velocity matrices that represent real-time traffic conditions at the intersection.	Spatio-temporal stacked occupancy representation. previous phase (N=10 steps)
action selection	Actions are selected by the PPO policy network, which generates a vector of control decisions, including signal timing adjustments, recommended route changes, and demand management strategies.	The agent selects the green light duration for the current signal phase based on the maximum Q-value output from the neural network. Action selection follows a phase-cycled structure.	two components: Intersection Action: Four predefined phase configurations, selection is drawn from a Categorical distribution; Mid-block Actions: For each of 7 signals, selects a binary action , each modeled as a Bernoulli distribution.
Action Scope	Network-wide control decisions including signal timing adjustments, route guidance recommendations, and demand management strategies designed to redistribute traffic demand and mitigate congestion.	The model adjusts only the green light duration of the current phase within a fixed cyclic phase sequence. The action set consists of predefined green durations. Single intersection	8 signals: 1 main intersection+ 7 mid-block signals

Agent Type	Centralized **single-agent** reinforcement learning controller managing **network**-wide traffic operations.	**Single-agent** reinforcement learning controller managing traffic signal timing at a **single** intersection, where one centralized agent observes the intersection traffic state and selects the optimal signal phase duration using a Deep Q-Network–based policy.	**Single-agent** reinforcement learning controller optimizing traffic signal timing for both pedestrian crossings and vehicle movements along an urban **corridor**, where a centralized agent makes phase control decisions based on combined pedestrian and vehicle traffic states.
reward design	Multi-objective (travel time, emissions, fairness, stability)	difference in total queue length between consecutive action time step	Exponentially Increasing Maximum Wait Aggregated Queue (EI-MWAQ); L2 aggregation + exponential penalty
DRL Structure	The decision optimization module follows an **Actor–Critic architecture using the PPO algorithm**, where a **policy network** generates traffic control actions and a **value network** estimates the value of network traffic states. The framework operates on graph-structured traffic data and integrates predictive traffic features generated by the GCRNN model.	The framework is based on an enhanced DQN architecture combining convolutional layers for traffic feature extraction and fully connected layers for Q-value estimation. It adopts dueling network decomposition, double Q-learning with main and target networks, prioritized experience replay, and noisy network parameters to improve convergence speed, stability, and robustness.	Algorithm: **PPO** Actor–Critic Architecture Separate MLP networks Hidden layers (Actor): [512, 256, 128, 64, 32] Hidden layers (Critic): [512, 256, 128, 64, 32]
Overestimation Fix	Not explicitly addressed in the paper. The model uses PPO (policy-gradient method), which does not rely on Q-value estimation like DQN-based approaches and therefore does not require overestimation correction mechanisms such as Double Q-learning.	using Double Q-learning, the dueling network architecture improves value estimation stability.	Not applicable (PPO policy-based method; no Q-learning overestimation)
Evaluation Metrics	Average travel time , Total Traffic Delay, Throughput , CO_2 Emissions , Fairness Index (Jain's fairness index) , System Responsiveness to Incident events.	Cumulative Reward per episode Average Waiting Time per lane (seconds) Average Queue Length per lane (vehicles)	Avg & total wait time (vehicle & pedestrian), conflict count, phase switching frequency, generalization under unseen demand.